RESUMO ÁREAS DE CONHECIMENTO SIPA: S-SUSTENTABILIDADE

REFERÊNCIA BÁSICA S-SUSTENTABILIDADE→AGENDA 2030/ONU

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BRASÍLIA, OUTUBRO/2026

1-DESENVOLVIMENTO SUSTENTÁVEL: AGENDA 2030 DA ONU;

2-INOVAÇÃO: OSLO MANUAL 2018 (OECD) E PESQUISA E DESENVOVIMENTO EXPERIMENTAL: FRASCATI MANUAL 2015 (OECD), E TECHNOLOGY READINESS LEVELS-TRLS: NASA;

3-PRIVACIDADE (LEI GERAL DE DADOS PESSOAIS-LGPD, BRASIL, GENERAL DATA PROTECTION REGULATION-GDPR, UNIÃO EUROPEIA, FAIR INFORMATION PRACTICE PRINCIPLES-FIPPS, EUA); E

4-ANÁLISE DE NEGÓCIOS COM DECISÕES AJUSTADAS A RISCOS (PMI GUIDE TO BUSINESS ANALYSIS/2017, ISO 31.000/2009, ISO 31.000/2018, ISO 31.010/2019, PMBOK6/2017).

PREMISSA BÁSICA

"QUALQUER **SISTEMA SOCIAL** (**"ORGANIZAÇÃO**", SEGUNDO JAGUARIBE) COM **PROPÓSITO**, QUE **APRENDE E ENSINA**, CARACTERIZADO COMO **SIPA** (**S**USTENTÁVEL **I**NOVADOR, **P**ROTETOR DA **P**RIVACIDADE, **A**NÁLISE DE NEGÓCIOS COM DECISÕES **A**JUSTADAS A RISCOS), DEVE TER COMO **PROPÓSITO** A **FORMAÇÃO "ÓTIMA" DE VALOR** (**VALORIZAÇÃO**) **E A APROPRIAÇÃO EQUÂNIME DO VALOR FORMADO** ÀS PARTES QUE PODEM AFETAR OU SER AFETADOS POR ESSA VALORIZAÇÃO (*STAKEHOLDERS*), SEJAM ELES INTERNOS OU EXTERNOS AO SISTEMA EM QUESTÃO, SEGUNDO **INDICADORES** DE **NÍVEIS** EM **TRAJETÓRIAS DE EXCELÊNCIA** (**EX1-**EFICIÊNCIA, **EX2-**EFICÁCIA, **EX3-**EMINÊNCIA, **EX4-**EFETIVIDADE) E DE **GRAUS** DE **EVOLUÇÃO ORGANIZACIONAL** (**EV1-**EXISTÊNCIA, **EV2-**EXPERIÊNCIA, **EV3-**EMERGÊNCIA, **EV4-**EVIDÊNCIA), **INDICADORES** ESSES ASSOCIADOS AOS **RESULTADOS** DE **TOMADA DE DECISÃO** (**ABSOLVER/reSOLVER/RESOLVER/DISSOLVER**) EM **MÚLTIPLOS NÍVEIS DE APRENDIZAGEM E ENSINO** (**FUNCIONAMENTO/PREMISSAS/INTELIGÊNCIA/CONTEXTOS**)".

(i) **17 ODS E AS 169 CORRESPONDENTES METAS DA 2030 AGENDA/UN**, (https://sustainabledevelopment.un.org/content/documents/21252030%20Agenda%20for%20Sustainable%20Development%20web.pdf);

(ii) **CONCEITOS DE INOVAÇÃO EM PRODUTOS, PROCESSOS E PRONTIDÃO DO OSLO MANUAL 2018/OECD** (http://www.oecd.org/science/oslo-manual-2018-9789264304604-en.htm), **CONCEITOS DE PESQUISA E DESENVOLVIMENTO EXPERIMENTAL DO FRASCATI MANUAL 2015/OECD** (https://www.oecd.org/en/publications/2015/10/frascati-manual-2015_g1g57dcb.html), **9 TECHNOLOGY READINESS LEVELS (TRLS) DA NASA** (https://www.nasa.gov/pdf/458490main_TRL_Definitions.pdf), **CIBERSEGURANÇA (CYBERSECURITY), COM ATENÇÃO NA EDUCAÇÃO EM CYBERSECURITY/US** (https://www.nist.gov/itl/applied-cybersecurity/nice);

(iii) **DISPOSITIVOS LEGAIS E SUAS REGULAMENTAÇÕES QUANTO À PROTEÇÃO DE DADOS PESSOAIS (LGPD/BRASIL:** HTTP://WWW.PLANALTO.GOV.BR/CCIVIL_03/_ATO2019-2022/2019/LEI/L13853.HTM, **GDPR/EU:** https://gdpr.eu/what-is-gdpr/**, PRIVACY ENGINEERING/US:** https://nvlpubs.nist.gov/nistpubs/ir/2017/NIST.IR.8062.pdf, https://blog.lukaszolejnik.com/privacy-engineering-principles-recommended-by-nist/);

(iv) **CONCEITOS BÁSICOS DE ANÁLISE DE NEGÓCIOS, SEGUNDO O PMI GUIDE TO BUSINESS ANALYSIS 2017 E DE GESTÃO DE RISCOS EM NORMAS ISO (ISO 31.000/2009, ISO 31.000/2018, ISO 31.010/2019)**.

SUSTENTABILIDADE E DESENVOLVIMENTO SUSTENTÁVEL: AGENDA 2030 DA ONU

UN 2030 AGENDA, (https://sustainabledevelopment.un.org/post2015/transformingourworld) CONTEMPLA 17 OBJETIVOS DE DESENVOLVIMENTO SUSTENTÁVEL – ODSS (EM INGLÊS, SUSTAINABLE DEVELOPMENT GOALS – SDGS). ACESSO À LISTA COMPLETA DOS 17 ODSS E SEUS RESPECTIVOS 169 ALVOS (TARGETS) EM INGLÊS: https://sdgs.un.org/goals E PORTUGUÊS: https://brasil.un.org/pt-br/sdgs.

1. **ERRADICAÇÃO DA POBREZA/NO POVERTY: ACABAR COM A POBREZA EM TODAS AS SUAS FORMAS, EM TODOS OS LUGARES/END POVERTY IN ALL ITS FORMS EVERYWHERE.**
2. **FOME ZERO E AGRICULTURA SUSTENTÁVEL/ZERO HUNGER: ACABAR COM A FOME, ALCANÇAR A SEGURANÇA ALIMENTAR E MELHORIA DA NUTRIÇÃO E PROMOVER A AGRICULTURA SUSTENTÁVEL/END HUNGER, ACHIEVE FOOD SECURITY AND IMPROVED NUTRITION AND PROMOTE SUSTAINABLE AGRICULTURE.**
3. **BOA SAÚDE E BEM-ESTAR/GOOD HEALTH AND WELL-BEING: ASSEGURAR UMA VIDA SAUDÁVEL E PROMOVER O BEM-ESTAR PARA TODOS, EM TODAS AS IDADES/ENSURE HEALTHY LIVES AND PROMOTE WELL-BEING FOR ALL AT ALL AGES.**
4. **EDUCAÇÃO DE QUALIDADE/QUALITY EDUCATION: ASSEGURAR A EDUCAÇÃO INCLUSIVE E EQUITATIVA E DE QUALIDADE, E PROMOVER OPORTUNIDADES DE APRENDIZAGEM AO LONGO DA VIDA PARA TODOS /ENSURE INCLUSIVE AND EQUITABLE QUALITY EDUCATION AND PROMOTE LIFELONG LEARNING OPPORTUNITIES FOR ALL.**
5. **IGUALDADE DE GÊNERO/GENDER EQUALITY: ALCANÇAR A IGUALDADE DE GÊNERO E EMPODERAR TODAS AS MULHERES E MENINAS/ACHIEVE GENDER EQUALITY AND EMPOWER ALL WOMEN AND GIRLS.**
6. **ÁGUA POTÁVEL E SANEAMENTO/CLEAN WATER AND SANITATION: ASSEGURAR A DISPONIBILIDADE E GESTÃO SUSTENTÁVEL DA ÁGUA E SANEAMENTO PARA TODOS/ENSURE AVAILABILITY AND SUSTAINABLE MANAGEMENT OF WATER AND SANITATION FOR ALL.**

S1.4.1.1-SUSTENTABILIDADE E DESENVOLVIMENTO SUSTENTÁVEL: AGENDA 2030 DA ONU

7. ENERGIA LIMPA E ACESSÍVEL/AFFORDABLE AND CLEAN ENERGY: ASSEGURAR O ACESSO CONFIÁVEL, SUSTENTÁVEL, MODERNO E A PREÇO ACESSÍVEL À ENERGIA PARA TODOS/ENSURE ACCESS TO AFFORDABLE, RELIABLE, SUSTAINABLE AND MODERN ENERGY FOR ALL.

8. TRABALHO DECENTE E CRESCIMENTO ECONÔMICO/DECENT WORK AND ECONOMIC GROWTH: PROMOVER O CRESCIMENTO ECONÔMICO SUSTENTADO, INCLUSIVO E SUSTENTÁVEL, EMPREGO PLENO E PRODUTIVO E TRABALHO DECENTE PARA TODOS/PROMOTE SUSTAINED, INCLUSIVE AND SUSTAINABLE ECONOMIC GROWTH, FULL AND PRODUCTIVE EMPLOYMENT AND DECENT WORK FOR ALL.

9. INDÚSTRIA, INOVAÇÃO E INFRAESTRUTURA/INDUSTRY, INNOVATION AND INFRASTRUCTURE: CONSTRUIR INFRAESTRUTURAS RESILIENTES, PROMOVER A INDUSTRIALIZAÇÃO INCLUSIVA E SUSTENTÁVEL E FOMENTAR A INOVAÇÃO/BUILD RESILIENT INFRASTRUCTURE, PROMOTE INCLUSIVE AND SUSTAINABLE INDUSTRIALIZATION AND FOSTER INNOVATION.

10. REDUÇÃO DAS DESIGUALDADES/REDUCED INEQUALITIES: REDUZIR A DESIGUALDADE DENTRO DOS PAÍSES E ENTRE ELES/REDUCE INEQUALITY WITHIN AND AMONG COUNTRIES.

11. CIDADES E COMUNIDADES SUSTENTÁVEIS/SUSTAINABLE CITIES AND COMMUNITIES: TORNAR AS CIDADES E OS ASSENTAMENTOS HUMANOS INCLUSIVOS, SEGUROS, RESILIENTES E SUSTENTÁVEIS/MAKE CITIES AND HUMAN SETTLEMENTS INCLUSIVE, SAFE, RESILIENT AND SUSTAINABLE.

12. CONSUMO E PRODUÇÃO RESPONSÁVEIS/RESPONSIBLE CONSUMPTION AND PRODUCTION: ASSEGURAR PADRÕES DE PRODUÇÃO E DE CONSUMO SUSTENTÁVEIS/ENSURE SUSTAINABLE CONSUMPTION AND PRODUCTION PATTERNS.

S1.4.1.1-SUSTENTABILIDADE E DESENVOLVIMENTO SUSTENTÁVEL: AGENDA 2030 DA ONU

13. AÇÃO CONTRA A MUDANÇA GLOBAL DO CLIMA/CLIMATE ACTION: TOMAR MEDIDAS URGENTES PARA COMBATER A MUDANÇA CLIMÁTICA E SEUS IMPACTOS (*RECONHECENDO QUE A CONVENÇÃO QUADRO DAS NAÇÕES UNIDAS SOBRE MUDANÇA DO CLIMA [UNFCCC] É O FÓRUM INTERNACIONAL INTERGOVERNAMENTAL PRIMÁRIO PARA NEGOCIAR A RESPOSTA GLOBAL À MUDANÇA DO CLIMA.)/TAKE URGENT ACTION TO COMBAT CLIMATE CHANGE AND ITS IMPACTS.

14. VIDA NA ÁGUA/LIFE BELOW WATER: CONSERVAÇÃO E USO SUSTENTÁVEL DOS OCEANOS, DOS MARES E DOS RECURSOS MARINHOS PARA O DESENVOLVIMENTO SUSTENTÁVE/CONSERVE AND SUSTAINABLY USE THE OCEANS, SEAS AND MARINE RESOURCES FOR SUSTAINABLE DEVELOPMENT.

15. VIDA TERRESTRE/LIFE ON LAND: PROTEGER, RECUPERAR E PROMOVER O USO SUSTENTÁVEL DOS ECOSSISTEMAS TERRESTRES, GERIR DE FORMA SUSTENTÁVEL AS FLORESTAS, COMBATER A DESERTIFICAÇÃO, DETER E REVERTER A DEGRADAÇÃO DA TERRA E DETER A PERDA DE BIODIVERSIDADE/PROTECT, RESTORE AND PROMOTE SUSTAINABLE USE OF TERRESTRIAL ECOSYSTEMS, SUSTAINABLY MANAGE FORESTS, COMBAT DESERTIFICATION, AND HALT AND REVERSE LAND DEGRADATION AND HALT BIODIVERSITY LOSS.

16. PAZ, JUSTIÇA E INSTITUIÇÕES EFICAZES/PEACE, JUSTICE AND STRONG INSTITUTIONS: PROMOVER SOCIEDADES PACÍFICAS E INCLUSIVAS PARA O DESENVOLVIMENTO SUSTENTÁVEL, PROPORCIONAR O ACESSO À JUSTIÇA PARA TODOS E CONSTRUIR INSTITUIÇÕES EFICAZES, RESPONSÁVEIS E INCLUSIVAS EM TODOS OS NÍVEIS /PROMOTE PEACEFUL AND INCLUSIVE SOCIETIES FOR SUSTAINABLE DEVELOPMENT, PROVIDE ACCESS TO JUSTICE FOR ALL AND BUILD EFFECTIVE, ACCOUNTABLE AND INCLUSIVE INSTITUTIONS AT ALL LEVELS.

17. PARCERIAS E MEIOS DE IMPLEMENTAÇÃO/PARTNERSHIPS FOR THE GOALS: FORTALECER OS MEIOS DE IMPLEMENTAÇÃO E REVITALIZAR A PARCERIA GLOBAL PARA O DESENVOLVIMENTO SUSTENTÁVEL /STRENGHTEN THE MEANS OF IMPLEMENTATION AND REVITALIZE THE GLOBAL PARTNERSHIP FOR SUSTAINABLE DEVELOPMENT.

SEGUE-SE A LISTA COMPLETA DOS 17 ODSS E RESPECTIVOS ALVOS (TARGETS) EM INGLÊS, QUE SERÃO OBJETO DE QUESTIONÁRIO

CÓPIA ONLINE: INGLÊS→ https://sdgs.un.org/goals E PORTUGUÊS→https://brasil.un.org/pt-br/sdgs

1. No poverty: end poverty in all its forms everywhere.

1.1 By 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than $1.25 a day

1.2 By 2030, reduce at least by half the proportion of men, women and children of all ages living in poverty in all its dimensions according to national definitions

1.3 Implement nationally appropriate social protection systems and measures for all, including floors, and by 2030 achieve substantial coverage of the poor and the vulnerable

1.4 By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance

1.5 By 2030, build the resilience of the poor and those in vulnerable situations and reduce their exposure and vulnerability to climate-related extreme events and other economic, social and environmental shocks and disasters

1.a Ensure significant mobilization of resources from a variety of sources, including through enhanced development cooperation, in order to provide adequate and predictable means for developing countries, in particular least developed countries, to implement programmes and policies to end poverty in all its dimensions

1.b Create sound policy frameworks at the national, regional and international levels, based on pro-poor and gender-sensitive development strategies, to support accelerated investment in poverty eradication actions

2. Zero hunger: end hunger, achieve food security and improved nutrition and promote sustainable agriculture.

2.1 By 2030, end hunger and ensure access by all people, in particular the poor and people in vulnerable situations, including infants, to safe, nutritious and sufficient food all year round

2.2 By 2030, end all forms of malnutrition, including achieving, by 2025, the internationally agreed targets on stunting and wasting in children under 5 years of age, and address the nutritional needs of adolescent girls, pregnant and lactating women and older persons

2.3 By 2030, double the agricultural productivity and incomes of small-scale food producers, in particular women, indigenous peoples, family farmers, pastoralists and fishers, including through secure and equal access to land, other productive resources and inputs, knowledge, financial services, markets and opportunities for value addition and non-farm employment

2.4 By 2030, ensure sustainable food production systems and implement resilient agricultural practices that increase productivity and production, that help maintain ecosystems, that strengthen capacity for adaptation to climate change, extreme weather, drought, flooding and other disasters and that progressively improve land and soil quality

2.5 By 2020, maintain the genetic diversity of seeds, cultivated plants and farmed and domesticated animals and their related wild species, including through soundly managed and diversified seed and plant banks at the national, regional and international levels, and promote access to and fair and equitable sharing of benefits arising from the utilization of genetic resources and associated traditional knowledge, as internationally agreed

2.a Increase investment, including through enhanced international cooperation, in rural infrastructure, agricultural research and extension services, technology development and plant and livestock gene banks in order to enhance agricultural productive capacity in developing countries, in particular least developed countries

2.b Correct and prevent trade restrictions and distortions in world agricultural markets, including through the parallel elimination of all forms of agricultural export subsidies and all export measures with equivalent effect, in accordance with the mandate of the Doha Development Round

2.c Adopt measures to ensure the proper functioning of food commodity markets and their derivatives and facilitate timely access to market information, including on food reserves, in order to help limit extreme food price volatility

3. Good health and well being: ensure healthy lives and promote well-being for all at all ages.
3.1 By 2030, reduce the global maternal mortality ratio to less than 70 per 100,000 live births
3.2 By 2030, end preventable deaths of newborns and children under 5 years of age, with all countries aiming to reduce neonatal mortality to at least as low as 12 per 1,000 live births and under-5 mortality to at least as low as 25 per 1,000 live births
3.3 By 2030, end the epidemics of AIDS, tuberculosis, malaria and neglected tropical diseases and combat hepatitis, water-borne diseases and other communicable diseases
3.4 By 2030, reduce by one third premature mortality from non-communicable diseases through prevention and treatment and promote mental health and well-being
3.5 Strengthen the prevention and treatment of substance abuse, including narcotic drug abuse and harmful use of alcohol
3.6 By 2020, halve the number of global deaths and injuries from road traffic accidents
3.7 By 2030, ensure universal access to sexual and reproductive health-care services, including for family planning, information and education, and the integration of reproductive health into national strategies and programmes
3.8 Achieve universal health coverage, including financial risk protection, access to quality essential health-care services and access to safe, effective, quality and affordable essential medicines and vaccines for all
3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination
3.a Strengthen the implementation of the World Health Organization Framework Convention on Tobacco Control in all countries, as appropriate
3.b Support the research and development of vaccines and medicines for the communicable and non-communicable diseases that primarily affect developing countries, provide access to affordable essential medicines and vaccines, in accordance with the Doha Declaration on the TRIPS Agreement and Public Health, which affirms the right of developing countries to use to the full the provisions in the Agreement on Trade-Related Aspects of Intellectual Property Rights regarding flexibilities to protect public health, and, in particular, provide access to medicines for all

4. Quality education: ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

4.1 By 2030, ensure that all girls and boys complete free, equitable and quality primary and secondary education leading to relevant and effective learning outcomes

4.2 By 2030, ensure that all girls and boys have access to quality early childhood development, care and pre-primary education so that they are ready for primary education

4.3 By 2030, ensure equal access for all women and men to affordable and quality technical, vocational and tertiary education, including university

4.4 By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship

4.5 By 2030, eliminate gender disparities in education and ensure equal access to all levels of education and vocational training for the vulnerable, including persons with disabilities, indigenous peoples and children in vulnerable situations

4.6 By 2030, ensure that all youth and a substantial proportion of adults, both men and women, achieve literacy and numeracy

4.7 By 2030, ensure that all learners acquire the knowledge and skills needed to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship and appreciation of cultural diversity and of culture's contribution to sustainable development

4.a Build and upgrade education facilities that are child, disability and gender sensitive and provide safe, non-violent, inclusive and effective learning environments for all

4.b By 2020, substantially expand globally the number of scholarships available to developing countries, in particular least developed countries, small island developing States and African countries, for enrolment in higher education, including vocational training and information and communications technology, technical, engineering and scientific programmes, in developed countries and other developing countries

4.c By 2030, substantially increase the supply of qualified teachers, including through international cooperation for teacher training in developing countries, especially least developed countries and small island developing States

5. Gender equality: achieve gender equality and empower all women and girls.
5.1 End all forms of discrimination against all women and girls everywhere
5.2 Eliminate all forms of violence against all women and girls in the public and private spheres, including trafficking and sexual and other types of exploitation
5.3 Eliminate all harmful practices, such as child, early and forced marriage and female genital mutilation
5.4 Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate
5.5 Ensure women's full and effective participation and equal opportunities for leadership at all levels of decision-making in political, economic and public life
5.6 Ensure universal access to sexual and reproductive health and reproductive rights as agreed in accordance with the Programme of Action of the International Conference on Population and Development and the Beijing Platform for Action and the outcome documents of their review conferences
5.a Undertake reforms to give women equal rights to economic resources, as well as access to ownership and control over land and other forms of property, financial services, inheritance and natural resources, in accordance with national laws
5.b Enhance the use of enabling technology, in particular information and communications technology, to promote the empowerment of women
5.c Adopt and strengthen sound policies and enforceable legislation for the promotion of gender equality and the empowerment of all women and girls at all levels

6. Clean water and sanitation: ensure availability and sustainable management of water and sanitation for all.
6.1 By 2030, achieve universal and equitable access to safe and affordable drinking water for all
6.2 By 2030, achieve access to adequate and equitable sanitation and hygiene for all and end open defecation, paying special attention to the needs of women and girls and those in vulnerable situations
6.3 By 2030, improve water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally
6.4 By 2030, substantially increase water-use efficiency across all sectors and ensure sustainable withdrawals and supply of freshwater to address water scarcity and substantially reduce the number of people suffering from water scarcity
6.5 By 2030, implement integrated water resources management at all levels, including through transboundary cooperation as appropriate
6.6 By 2020, protect and restore water-related ecosystems, including mountains, forests, wetlands, rivers, aquifers and lakes
6.a By 2030, expand international cooperation and capacity-building support to developing countries in water- and sanitation-related activities and programmes, including water harvesting, desalination, water efficiency, wastewater treatment, recycling and reuse technologies
6.b Support and strengthen the participation of local communities in improving water and sanitation management
7. Affordable and clean energy: ensure access to affordable, reliable, sustainable and modern energy for all.
7.1 By 2030, ensure universal access to affordable, reliable and modern energy services
7.2 By 2030, increase substantially the share of renewable energy in the global energy mix
7.3 By 2030, double the global rate of improvement in energy efficiency
7.a By 2030, enhance international cooperation to facilitate access to clean energy research and technology, including renewable energy, energy efficiency and advanced and cleaner fossil-fuel technology, and promote investment in energy infrastructure and clean energy technology
7.b By 2030, expand infrastructure and upgrade technology for supplying modern and sustainable energy services for all in developing countries, in particular least developed countries, small island developing States, and land-locked developing countries, in accordance with their respective programmes of support

8. Decent work and economic growth: promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

8.1 Sustain per capita economic growth in accordance with national circumstances and, in particular, at least 7 per cent gross domestic product growth per annum in the least developed countries

8.2 Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors

8.3 Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services

8.4 Improve progressively, through 2030, global resource efficiency in consumption and production and endeavour to decouple economic growth from environmental degradation, in accordance with the 10-year framework of programmes on sustainable consumption and production, with developed countries taking the lead

8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value

8.6 By 2020, substantially reduce the proportion of youth not in employment, education or training

8.7 Take immediate and effective measures to eradicate forced labour, end modern slavery and human trafficking and secure the prohibition and elimination of the worst forms of child labour, including recruitment and use of child soldiers, and by 2025 end child labour in all its forms

8.8 Protect labour rights and promote safe and secure working environments for all workers, including migrant workers, in particular women migrants, and those in precarious employment

8.9 By 2030, devise and implement policies to promote sustainable tourism that creates jobs and promotes local culture and products

8.10 Strengthen the capacity of domestic financial institutions to encourage and expand access to banking, insurance and financial services for all

8a E 8b→INÍCIO PRÓXIMA PÁGINA

8.a Increase Aid for Trade support for developing countries, in particular least developed countries, including through the Enhanced Integrated Framework for Trade-Related Technical Assistance to Least Developed Countries
8.b By 2020, develop and operationalize a global strategy for youth employment and implement the Global Jobs Pact of the International Labour Organization

9. Industry, innovation and infrastructure: promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

9.1 Develop quality, reliable, sustainable and resilient infrastructure, including regional and transborder infrastructure, to support economic development and human well-being, with a focus on affordable and equitable access for all
9.2 Promote inclusive and sustainable industrialization and, by 2030, significantly raise industry's share of employment and gross domestic product, in line with national circumstances, and double its share in least developed countries
9.3 Increase the access of small-scale industrial and other enterprises, in particular in developing countries, to financial services, including affordable credit, and their integration into value chains and markets
9.4 By 2030, upgrade infrastructure and retrofit industries to make them sustainable, with increased resource-use efficiency and greater adoption of clean and environmentally sound technologies and industrial processes, with all countries taking action in accordance with their respective capabilities
9.5 Enhance scientific research, upgrade the technological capabilities of industrial sectors in all countries, in particular developing countries, including, by 2030, encouraging innovation and substantially increasing the number of research and development workers per 1 million people and public and private research and development spending
9.a Facilitate sustainable and resilient infrastructure development in developing countries through enhanced financial, technological and technical support to African countries, least developed countries, landlocked developing countries and small island developing States
9.b Support domestic technology development, research and innovation in developing countries, including by ensuring a conducive policy environment for, inter alia, industrial diversification and value addition to commodities
9.c Significantly increase access to information and communications technology and strive to provide universal and affordable access to the Internet in least developed countries by 2020

10. Reduced inequalities: reduce inequality within and among countries.
10.1 By 2030, progressively achieve and sustain income growth of the bottom 40 per cent of the population at a rate higher than the national average
10.2 By 2030, empower and promote the social, economic and political inclusion of all, irrespective of age, sex, disability, race, ethnicity, origin, religion or economic or other status
10.3 Ensure equal opportunity and reduce inequalities of outcome, including by eliminating discriminatory laws, policies and practices and promoting appropriate legislation, policies and action in this regard
10.4 Adopt policies, especially fiscal, wage and social protection policies, and progressively achieve greater equality
10.5 Improve the regulation and monitoring of global financial markets and institutions and strengthen the implementation of such regulations
10.6 Ensure enhanced representation and voice for developing countries in decision-making in global international economic and financial institutions in order to deliver more effective, credible, accountable and legitimate institutions
10.7 Facilitate orderly, safe, regular and responsible migration and mobility of people, including through the implementation of planned and well-managed migration policies
10.a Implement the principle of special and differential treatment for developing countries, in particular least developed countries, in accordance with World Trade Organization agreements
10.b Encourage official development assistance and financial flows, including foreign direct investment, to States where the need is greatest, in particular least developed countries, African countries, small island developing States and landlocked developing countries, in accordance with their national plans and programmes
10.c By 2030, reduce to less than 3 per cent the transaction costs of migrant remittances and eliminate remittance corridors with costs higher than 5 per cent

11. Sustainable cities and communities: make cities and human settlements inclusive, safe, resilient and sustainable.

11.1 By 2030, ensure access for all to adequate, safe and affordable housing and basic services and upgrade slums

11.2 By 2030, provide access to safe, affordable, accessible and sustainable transport systems for all, improving road safety, notably by expanding public transport, with special attention to the needs of those in vulnerable situations, women, children, persons with disabilities and older persons

11.3 By 2030, enhance inclusive and sustainable urbanization and capacity for participatory, integrated and sustainable human settlement planning and management in all countries

11.4 Strengthen efforts to protect and safeguard the world's cultural and natural heritage

11.5 By 2030, significantly reduce the number of deaths and the number of people affected and substantially decrease the direct economic losses relative to global gross domestic product caused by disasters, including water-related disasters, with a focus on protecting the poor and people in vulnerable situations

11.6 By 2030, reduce the adverse per capita environmental impact of cities, including by paying special attention to air quality and municipal and other waste management

11.7 By 2030, provide universal access to safe, inclusive and accessible, green and public spaces, in particular for women and children, older persons and persons with disabilities

11.a Support positive economic, social and environmental links between urban, peri-urban and rural areas by strengthening national and regional development planning

11.b By 2020, substantially increase the number of cities and human settlements adopting and implementing integrated policies and plans towards inclusion, resource efficiency, mitigation and adaptation to climate change, resilience to disasters, and develop and implement, in line with the Sendai Framework for Disaster Risk Reduction 2015-2030, holistic disaster risk management at all levels

11.c Support least developed countries, including through financial and technical assistance, in building sustainable and resilient buildings utilizing local materials

12. Responsible consumption and production: ensure sustainable consumption and production patterns.

12.1 Implement the 10-year framework of programmes on sustainable consumption and production, all countries taking action, with developed countries taking the lead, taking into account the development and capabilities of developing countries

12.2 By 2030, achieve the sustainable management and efficient use of natural resources

12.3 By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses

12.4 By 2020, achieve the environmentally sound management of chemicals and all wastes throughout their life cycle, in accordance with agreed international frameworks, and significantly reduce their release to air, water and soil in order to minimize their adverse impacts on human health and the environment

12.5 By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse

12.6 Encourage companies, especially large and transnational companies, to adopt sustainable practices and to integrate sustainability information into their reporting cycle

12.7 Promote public procurement practices that are sustainable, in accordance with national policies and priorities

12.8 By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature

12.a Support developing countries to strengthen their scientific and technological capacity to move towards more sustainable patterns of consumption and production

12.b Develop and implement tools to monitor sustainable development impacts for sustainable tourism that creates jobs and promotes local culture and products

12.c Rationalize inefficient fossil-fuel subsidies that encourage wasteful consumption by removing market distortions, in accordance with national circumstances, including by restructuring taxation and phasing out those harmful subsidies, where they exist, to reflect their environmental impacts, taking fully into account the specific needs and conditions of developing countries and minimizing the possible adverse impacts on their development in a manner that protects the poor and the affected communities

13. Climate action: take urgent action to combat climate change and its impacts.

13.1 Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters in all countries

13.2 Integrate climate change measures into national policies, strategies and planning

13.3 Improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning

13.a Implement the commitment undertaken by developed-country parties to the United Nations Framework Convention on Climate Change to a goal of mobilizing jointly $100 billion annually by 2020 from all sources to address the needs of developing countries in the context of meaningful mitigation actions and transparency on implementation and fully operationalize the Green Climate Fund through its capitalization as soon as possible

13.b Promote mechanisms for raising capacity for effective climate change-related planning and management in least developed countries and small island developing States, including focusing on women, youth and local and marginalized communities

* Acknowledging that the United Nations Framework Convention on Climate Change is the primary international, intergovernmental forum for negotiating the global response to climate change.

14. Life below water: conserve and sustainably use the oceans, seas and marine resources for sustainable development.

14.1 By 2025, prevent and significantly reduce marine pollution of all kinds, in particular from land-based activities, including marine debris and nutrient pollution

14.2 By 2020, sustainably manage and protect marine and coastal ecosystems to avoid significant adverse impacts, including by strengthening their resilience, and take action for their restoration in order to achieve healthy and productive oceans

14.3 Minimize and address the impacts of ocean acidification, including through enhanced scientific cooperation at all levels

14.4 By 2020, effectively regulate harvesting and end overfishing, illegal, unreported and unregulated fishing and destructive fishing practices and implement science-based management plans, in order to restore fish stocks in the shortest time feasible, at least to levels that can produce maximum sustainable yield as determined by their biological characteristics

14.5 By 2020, conserve at least 10 per cent of coastal and marine areas, consistent with national and international law and based on the best available scientific information

14.6 By 2020, prohibit certain forms of fisheries subsidies which contribute to overcapacity and overfishing, eliminate subsidies that contribute to illegal, unreported and unregulated fishing and refrain from introducing new such subsidies, recognizing that appropriate and effective special and differential treatment for developing and least developed countries should be an integral part of the World Trade Organization fisheries subsidies negotiation

14.7 By 2030, increase the economic benefits to Small Island developing States and least developed countries from the sustainable use of marine resources, including through sustainable management of fisheries, aquaculture and tourism

14.a Increase scientific knowledge, develop research capacity and transfer marine technology, taking into account the Intergovernmental Oceanographic Commission Criteria and Guidelines on the Transfer of Marine Technology, in order to improve ocean health and to enhance the contribution of marine biodiversity to the development of developing countries, in particular small island developing States and least developed countries

14.b Provide access for small-scale artisanal fishers to marine resources and markets

14.c Enhance the conservation and sustainable use of oceans and their resources by implementing international law as reflected in UNCLOS, which provides the legal framework for the conservation and sustainable use of oceans and their resources, as recalled in paragraph 158 of The Future We Want

15. Life on land: protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.

15.1 By 2020, ensure the conservation, restoration and sustainable use of terrestrial and inland freshwater ecosystems and their services, in particular forests, wetlands, mountains and drylands, in line with obligations under international agreements

15.2 By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase afforestation and reforestation globally

15.3 By 2030, combat desertification, restore degraded land and soil, including land affected by desertification, drought and floods, and strive to achieve a land degradation-neutral world

15.4 By 2030, ensure the conservation of mountain ecosystems, including their biodiversity, in order to enhance their capacity to provide benefits that are essential for sustainable development

15.5 Take urgent and significant action to reduce the degradation of natural habitats, halt the loss of biodiversity and, by 2020, protect and prevent the extinction of threatened species

15.6 Promote fair and equitable sharing of the benefits arising from the utilization of genetic resources and promote appropriate access to such resources, as internationally agreed

15.7 Take urgent action to end poaching and trafficking of protected species of flora and fauna and address both demand and supply of illegal wildlife products

15.8 By 2020, introduce measures to prevent the introduction and significantly reduce the impact of invasive alien species on land and water ecosystems and control or eradicate the priority species

15.9 By 2020, integrate ecosystem and biodiversity values into national and local planning, development processes, poverty reduction strategies and accounts

15a, 15b E 15c→INÍCIO PRÓXIMA PÁGINA

15.a Mobilize and significantly increase financial resources from all sources to conserve and sustainably use biodiversity and ecosystems
15.b Mobilize significant resources from all sources and at all levels to finance sustainable forest management and provide adequate incentives to developing countries to advance such management, including for conservation and reforestation
15.c Enhance global support for efforts to combat poaching and trafficking of protected species, including by increasing the capacity of local communities to pursue sustainable livelihood opportunities

16. Peace, justice and strong institutions: promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels.

16.1 Significantly reduce all forms of violence and related death rates everywhere
16.2 End abuse, exploitation, trafficking and all forms of violence against and torture of children
16.3 Promote the rule of law at the national and international levels and ensure equal access to justice for all
16.4 By 2030, significantly reduce illicit financial and arms flows, strengthen the recovery and return of stolen assets and combat all forms of organized crime
16.5 Substantially reduce corruption and bribery in all their forms
16.6 Develop effective, accountable and transparent institutions at all levels
16.7 Ensure responsive, inclusive, participatory and representative decision-making at all levels
16.8 Broaden and strengthen the participation of developing countries in the institutions of global governance
16.9 By 2030, provide legal identity for all, including birth registration
16.10 Ensure public access to information and protect fundamental freedoms, in accordance with national legislation and international agreements
16.a Strengthen relevant national institutions, including through international cooperation, for building capacity at all levels, in particular in developing countries, to prevent violence and combat terrorism and crime
16.b Promote and enforce non-discriminatory laws and policies for sustainable development

17. Partnerships for the goals: strengthen the means of implementation and revitalize the global partnership for sustainable development.

Finance

17.1 Strengthen domestic resource mobilization, including through international support to developing countries, to improve domestic capacity for tax and other revenue collection

17.2 Developed countries to implement fully their official development assistance commitments, including the commitment by many developed countries to achieve the target of 0.7 per cent of ODA/GNI to developing countries and 0.15 to 0.20 per cent of ODA/GNI to least developed countries; ODA providers are encouraged to consider setting a target to provide at least 0.20 per cent of ODA/GNI to least developed countries

17.3 Mobilize additional financial resources for developing countries from multiple sources

17.4 Assist developing countries in attaining long-term debt sustainability through coordinated policies aimed at fostering debt financing, debt relief and debt restructuring, as appropriate, and address the external debt of highly indebted poor countries to reduce debt distress

17.5 Adopt and implement investment promotion regimes for least developed countries

Technology

17.6 Enhance North-South, South-South and triangular regional and international cooperation on and access to science, technology and innovation and enhance knowledge sharing on mutually agreed terms, including through improved coordination among existing mechanisms, in particular at the United Nations level, and through a global technology facilitation mechanism

17.7 Promote the development, transfer, dissemination and diffusion of environmentally sound technologies to developing countries on favourable terms, including on concessional and preferential terms, as mutually agreed

17.8 Fully operationalize the technology bank and science, technology and innovation capacity-building mechanism for least developed countries by 2017 and enhance the use of enabling technology, in particular information and communications technology

17. Partnerships for the goals: strengthen the means of implementation and revitalize the global partnership for sustainable development.

Capacity-building

17.9 Enhance international support for implementing effective and targeted capacity-building in developing countries to support national plans to implement all the sustainable development goals, including through North-South, South-South and triangular cooperation

Trade

17.10 Promote a universal, rules-based, open, non-discriminatory and equitable multilateral trading system under the World Trade Organization, including through the conclusion of negotiations under its Doha Development Agenda

17.11 Significantly increase the exports of developing countries, in particular with a view to doubling the least developed countries' share of global exports by 2020

17.12 Realize timely implementation of duty-free and quota-free market access on a lasting basis for all least developed countries, consistent with World Trade Organization decisions, including by ensuring that preferential rules of origin applicable to imports from least developed countries are transparent and simple, and contribute to facilitating market access

17. Partnerships for the goals: strengthen the means of implementation and revitalize the global partnership for sustainable development.

Systemic issues

Policy and institutional coherence

17.13 Enhance global macroeconomic stability, including through policy coordination and policy coherence

17.14 Enhance policy coherence for sustainable development

17.15 Respect each country's policy space and leadership to establish and implement policies for poverty eradication and sustainable development

Multi-stakeholder partnerships

17.16 Enhance the global partnership for sustainable development, complemented by multi-stakeholder partnerships that mobilize and share knowledge, expertise, technology and financial resources, to support the achievement of the sustainable development goals in all countries, in particular developing countries

17.17 Encourage and promote effective public, public-private and civil society partnerships, building on the experience and resourcing strategies of partnerships

Data, monitoring and accountability

17.18 By 2020, enhance capacity-building support to developing countries, including for least developed countries and small island developing States, to increase significantly the availability of high-quality, timely and reliable data disaggregated by income, gender, age, race, ethnicity, migratory status, disability, geographic location and other characteristics relevant in national contexts

17.19 By 2030, build on existing initiatives to develop measurements of progress on sustainable development that complement gross domestic product, and support statistical capacity-building in developing countries

Ps DE DESENVOLVIMENTO SUSTENTÁVEL: MODELO DE TRIÂNGULOS E TETRAEDROS

A FIGURA AO LADO EXPLICITA O TRIÂNGULO ASSOCIADO AO DESENVOLVIMENTO SUSTENTÁVEL COMO SÍNTESE DE COMPONENTES AMBIENTAIS, ECONÔMICOS E SOCIAIS.



SEGUNDO A ONU, OS 17 OBJETIVOS DE DESENVOLVIMENTO SUSTENTÁVEL – ODSS (SUSTAINABLE DEVELOPMENT GOALS – SDGS) DA AGENDA 2030 DA ONU PODEM SER ALOCADOS A 5 PS DA SEGUINTE FORMA:

- PEOPLE (PESSOAS): SDG1-NO POVERTY; SDG2-ZERO HUNGER; SDG3-GOOD HEALTH AND WELL BEING; SDG4-QUALITY EDUCATION; SDG5-GENDER EQUALITY.
- PLANET (PLANETA): SDG6-CLEAN WATER AND SANITATION; SDG12-RESPONSIBLE CONSUMPTION AND PRODUTION; SDG13-CLIMATE ACTION; SDG14-LIFE BELLOW WATER; SDG15-LIFE ON LAND.
- PROSPERITY (PROSPERIDADE): SDG7-AFFORDABLE AND CLEAN ENERGY; SDG8-DECENT WORK AND AND ECONOMIC GROWTH; SDG9-INDUSTRY, INNOVATION AND INFRASTRUCTURE; SDG10-REDUCE INEQUALITIES; SDG11-SUSTAINABLE CITIES AND COMMUNITIES.
- PEACE (PAZ): SDG16-PEACE, JUSTICE AND STRONG INSTITUTIONS.
- PARTNERSHIP (PARCERIA): SDG17-PARTNERSHIPS AND GOALS.

COMO DECORRÊNCIA DESSA ALOCAÇÃO DOS 17 ODSS (SDGS) AOS 5 PS, AS 169 METAS (TARGETS) RELATIVAS AOS 17 ODSS TAMBÉM FICAM VINCULADAS A UM DOS 5 PS DE ALOCAÇÃO DO RESPECTIVO ODS. ASSIM, OUTRA MANEIRA DE VER O DESENVOLVIMENTO SUSTENTÁVEL É COLOCAR A ATENÇÃO NOS 5 PS E NÃO DIRETAMENTE NOS ASPECTOS AMBIENTAIS, ECONÔMICOS E SOCIAIS.

CONSIDERANDO A PROXIMIDADE DE PAZ (PEACE) E PARCERIA (PARTNERSHIP), PODE-SE CONSIDERAR A FUSÃO DELES, DANDO ORIGEM AO P DE PROVISÃO RESULTANTE DE PARCERIA COM PAZ (PROVISION AS RESULT OF PARTNERSHIP WITH PEACE). O QUADRO EXPLICITA A CONSOLIDAÇÃO EM 4 PS.

SUSTAINABLE DEVELOPMENT

- **TRIANGLES:**
  - **PEOPLE(SDGs 1,2,3,4,5):PPE=α=(α1,α2,α3)**
    - **α1=SURVIVAL; α2=WELL BEING; α3=QUALITY EDUCTION&GENDER EQUALITY**
  - **PLANET(SGs 6,12,13,14,15):PNT=β=(β1,β2,β3)**
    - **β1=LITHOSPHERE; β2=HYDROSPHERE; β3=ATMOSPHERE**
  - **PROSPERITY(SDGs 7,8,9,10,11):PRT=Y=(Y1,Y2,Y3)**
    - **Y1=PRODUCTION&CONSUMPTION; Y2=SUST CITIES&COMMUNIT; Y3=DEC W&RED INEQ**
  - **PROVISION(SDGs 16,17):PRV=δ=(δ1,δ2,δ3)**
    - **δ1=PEACE&JUSTICE; δ2=STRONG INSTITUTIONS; δ3=PARTNERSHIPS**

PARA FINS DE BUSCAS BIBLIOMÉTRICAS, OS RELACIONAMENTOS ENTRE OS QUATRO **PS** DE **DESENVOLVIMENTO SUSTENTÁVEL**, PODEM SER MODELADOS EM TERMOS DE **TRIÂNGULOS** E **TETRAEDROS**.

NO ÂMBITO DA ABORDAGEM DE TRIÂNGULOS E TETRAEDROS, O PRIMEIRO PASSO É CONSOLIDAR OS **5 PS** EM 4 CONCENTRAÇÕES, DENOTADAS PELAS LETRAS GREGAS **A, B, Y, Δ**: (**I**) **P1=PESSOAS=PEOPLE=PPE=A**; (**II**) **P2=PLANETA=PLANETPNT=B**; (**III**) **P3=PROSPERIDADE=PROSPERITY=PRT=Y**; E **P4=PROVISÃO COM PARCERIA E PAZ=PROVISION WITH PARTNERSHIP AND PEACE=PRV=Δ**, EM QUE **PROVISÃO COM PARCERIA E PAZ=PROVISION WITH PARTNERSHIP AND PEACE** É RESULTADO DA CONCENTRAÇÃO DE **2 PS**: **PEACE** AND **PARTNERSHIP**. O QUADRO ESTRUTURA OS COMPONENTES DO **DESENVOLVIMENTO SUSTENTÁVEL** EM TERMOS DESSES **4 PS**, COM DESTAQUE PARA "**PESSOAS=PEOPLE**".

UN 2030 AGENDA→CONCENTRATIONS OF SUSTAINABLE DEVELOPMENT GOALS (SDGs): SDEV		
PEOPLE(SDGs 1,2,3,4,5) α=PPE		
SURVIVAL α1=SRV(1,2)=PPE1		
WELL BEING α2=BEN(3)=PPE2		
QUALITY EDUCATION&GENDER EQUALITY α3=QEG(4,5)=PPE3		
PLANET(SDGs 6,12,13,14,15) β=PNT	**PROSPERITY(SDGs 7,8,9,10,11) Y=PRT**	**PROVISION(SDGs 16,17) δ=PRV**
LITHOSPHERE β1=LIT=PNT1	**RESP PROD&COM Y1=PDC=PRT1**	**PEACE AND JUSTICE δ1=PJU=PRV1**
HYDROSPHERE β2=HYD=PNT2	**SUST CITIES&COMM Y2=SCC=PRT2**	**STRONG INSTITUTIONS δ2=SIN=PRV2**
ATMOSPHERE β3=ATM=PNT3	**DEC WORK&RED INEQ Y3=DRI=PRT3**	**PARTNERSHIP δ3=PAR=PRV3**

COMO BIBLIOMETRIA EM IEPR É ESSENCIALMENTE A UTILIZAÇÃO DE ESTATÍSTICA EM BASES DE DADOS DE PUBLICAÇÕES CIENTÍFICAS, VALE DESTACAR OS CONCEITOS BÁSICOS DE ANÁLISE EXPLORATÓRIA DE DADOS/EXPLORATORY DATA ANALYSIS.

EM UMA ABORDAGEM ESSENCIALMENTE PRÁTICA, MOORE, DAVID S., ET AL. (A PRÁTICA DA ESTATÍSTICA EMPRESARIAL – COMO USAR DADOS PARA TOMAR DECISÕES, LTC, 2006, P. 5-6) REFEREM-SE À ANÁLISE EXPLORATÓRIA DE DADOS COMO "AS FERRAMENTAS E AS IDEIAS ESTATÍSTICAS QUE NOS AJUDAM A EXAMINAR OS DADOS COM O PROPÓSITO DE DESCREVER AS SUAS CARACTERÍSTICAS PRINCIPAIS". EM SEGUIDA DESCREVEM 2 ESTRATÉGIAS PARA A EXPLORAÇÃO ANALÍTICA DE UM CONJUNTO DE DADOS: (I) "COMECE EXAMINANDO CADA VARIÁVEL POR SI MESMA. EM SEGUIDA, PASSE A ESTUDAR AS RELAÇÕES EXISTENTES ENTRE AS VARIÁVEIS; (II) COMECE FAZENDO UM OU MAIS GRÁFICOS. FAÇA TAMBÉM RESUMOS NUMÉRICOS DE ASPECTOS ESPECÍFICOS DOS DADOS".

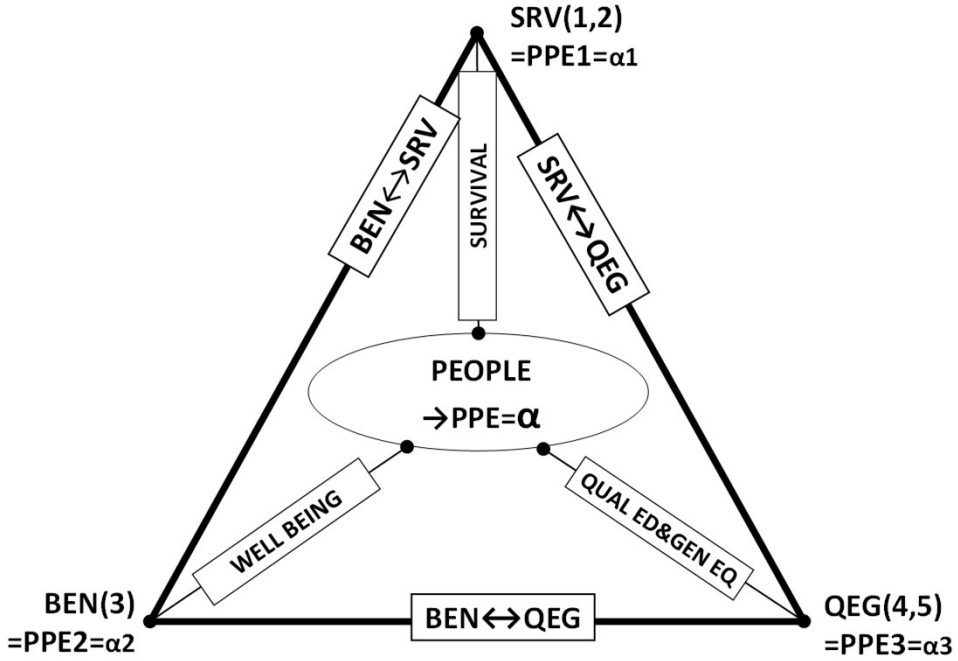
SEGUINDO A MESMA LINHA, ANDERSEN, DAVID R., ET AL. (ESTATÍSTICA APLICADA À ADMINISTRAÇÃO E ECONOMIA, THOMSOM, 2ª ED., 2007, P. 38) ESCREVEM QUE "AS TÉCNICAS DE ANÁLISE EXPLORATÓRIA DE DADOS CONSISTEM EM CÁLCULOS ARITMÉTICOS SIMPLES E EM GRÁFICOS FÁCEIS DE DESENHAR QUE PODEM SER USADOS PARA SINTETIZAR DADOS RAPIDAMENTE".

SEGUINDO UMA ABORDAGEM MAIS ROBUSTA CONCEITUALMENTE, O IBM CLOUD LEARN HUB (HTTPS://WWW.IBM.COM/BR-PT/CLOUD/LEARN/EXPLORATORY-DATA-ANALYSIS, ACESSO 13/08/2022), ESTABELECE QUE:

> **"A ANÁLISE EXPLORATÓRIA DE DADOS (EDA) É USADA POR CIENTISTAS DE DADOS PARA ANALISAR E INVESTIGAR CONJUNTOS DE DADOS E RESUMIR SUAS PRINCIPAIS CARACTERÍSTICAS, MUITAS VEZES USANDO MÉTODOS DE VISUALIZAÇÃO DE DADOS. ELA PERMITE DETERMINAR A MELHOR FORMA DE CONTROLAR AS FONTES DE DADOS PARA OBTER AS RESPOSTAS QUE VOCÊ PRECISA, TORNANDO MAIS FÁCIL PARA OS CIENTISTAS DE DADOS DESCOBRIREM PADRÕES, DETECTAREM ANOMALIAS, TESTAREM UMA HIPÓTESE OU VERIFICAREM SUPOSIÇÕES". EM COMPLEMENTO, EXPLICITA QUE O "OBJETIVO PRINCIPAL DA EDA É AJUDAR A ANALISAR DADOS ANTES DE FAZER QUALQUER SUPOSIÇÃO. ELA PODE AJUDAR NA IDENTIFICAÇÃO DE ERROS ÓBVIOS E ENTENDER MELHOR OS PADRÕES PRESENTES DOS DADOS, DETECTAR DESVIOS OU EVENTOS ANÔMALOS, ALÉM DE ENCONTRAR RELAÇÕES INTERESSANTES ENTRE AS VARIÁVEIS.**
>
> **OS CIENTISTAS DE DADOS PODEM USAR A ANÁLISE EXPLORATÓRIA PARA ASSEGURAR QUE OS RESULTADOS PRODUZIDOS SEJAM VÁLIDOS E APLICÁVEIS A QUAISQUER RESULTADOS E METAS DE NEGÓCIOS DESEJADOS. A EDA TAMBÉM AJUDA AS PARTES INTERESSADAS ATRAVÉS DA CONFIRMAÇÃO DE QUE ESTÃO FAZENDO AS PERGUNTAS CERTAS. ELA PODE AJUDAR A RESPONDER PERGUNTAS SOBRE DESVIOS PADRÃO, VARIÁVEIS CATEGÓRICAS E INTERVALOS DE CONFIANÇA. UMA VEZ QUE A EDA ESTIVER COMPLETA E OS INSIGHTS ESTEJAM DEFINIDOS, SEUS RECURSOS PODEM ENTÃO SER USADOS PARA MODELAGEM OU ANÁLISE DE DADOS MAIS SOFISTICADA, INCLUINDO MACHINE LEARNING."**

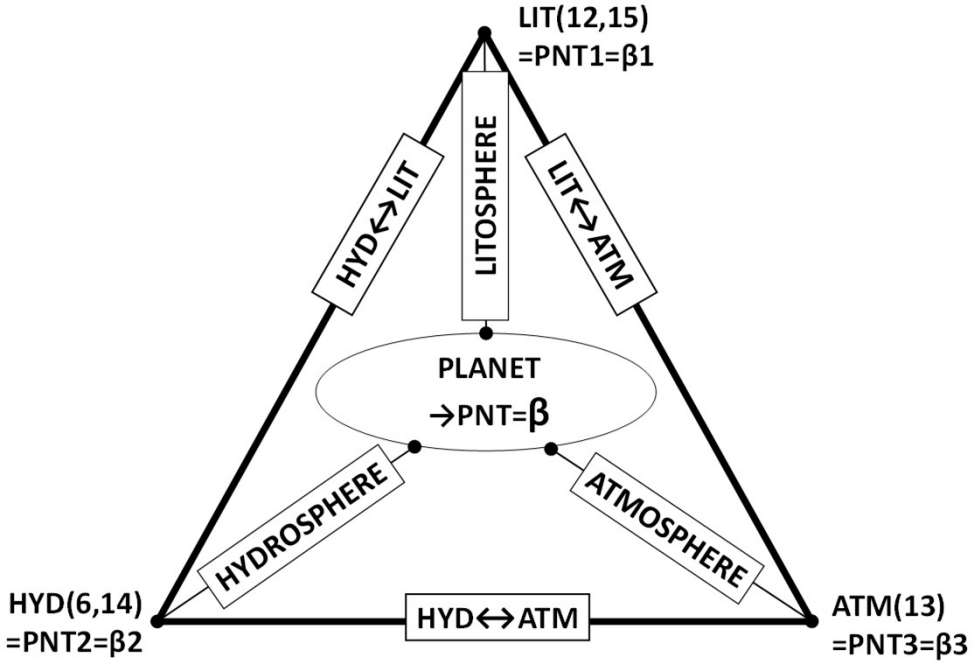
UMA DAS MAIS COMPLETAS PUBLICAÇÕES EM ESTATÍSTICA APLICADA À ENGENHARIA, O NIST/SEMATECH E-HANDBOOK OF STATISTICAL METHODS (https://www.itl.nist.gov/div898/handbook/, ACESSO 13/08/2022) RESERVA O PRIMEIRO DE SEUS 8 CAPÍTULOS PARA EXPLORATORY DATA ANALYSIS, CONCEITUADA COMO "AN APPROACH/PHILOSOPHY FOR DATA ANALYSIS THAT EMPLOYS A VARIETY OF TECHNIQUES (MOSTLY GRAPHICAL) TO: 1-MAXIMIZE INSIGHT INTO S DATA SET; 2-UNCOVER UNDERLYING STRUCTURE; 3-EXTRACT IMPORTANTE VARIABLES; 4-DETECT OUTLIERS AND ANOMALIES; 5-TEST UNDERLYING ASSUMPTIONS; 6-DEVELOP PARSIMONIUS MODELS; AND 7-DETERMINE OPTIMAL FATOR SETTINGS". RECOMENDA-SE O MANUAL NIST COMO REFERÊNCIA PRIMÁRIA PARA OS TRABALHOS EM EQUIPE.

SEGUNDO A MODELAGEM EM TRIÂNGULOS E TETRAEDROS, O TRIÂNGULO DA FIGURA MOSTRA A CONSOLIDAÇÃO DE P=PPE=PESSOA=PEOPLE=α (ODSs 1,2,3,4,5) EM TERMOS DE SRV(1,2)=PPE1=α1=SURVIVAL (ODSs 1,2); BEN(3)=PPE2=α2=WELL BEING (ODS 3); QEG(4,5)=PPE3=α3=QUALITY EDUCATION & GENDER EQUALITY (ODSs 4,5). ALÉM DE CONFIGURAR PEOPLE=PPE=α COMO SÍNTESE DOS 3 COMPONENTES EM SUA FACE, AS ARESTAS DO TRIÂNGULO DA FIGURA MOSTRAM OS RELACIONAMENTOS DIRECIONADOS ENTRE SEUS COMPONENTES. POR EXEMPLO, BEN→QEG ENFATIZA RELACIONAMENTOS DE ASPECTOS PRIMARIAMENTE DA CONCENTRAÇÃO DIRECIONADOS A ASPECTOS DA CONCENTRAÇÃO QEG=α3, ENQUANTO BEN←QEG DIZ RESPEITO A RELACIONAMENTOS ESSENCIALMENTE DA CONCENTRAÇÃO QEG=α3 DIRECIONADOS À CONCENTRAÇÃO BEN=α2.



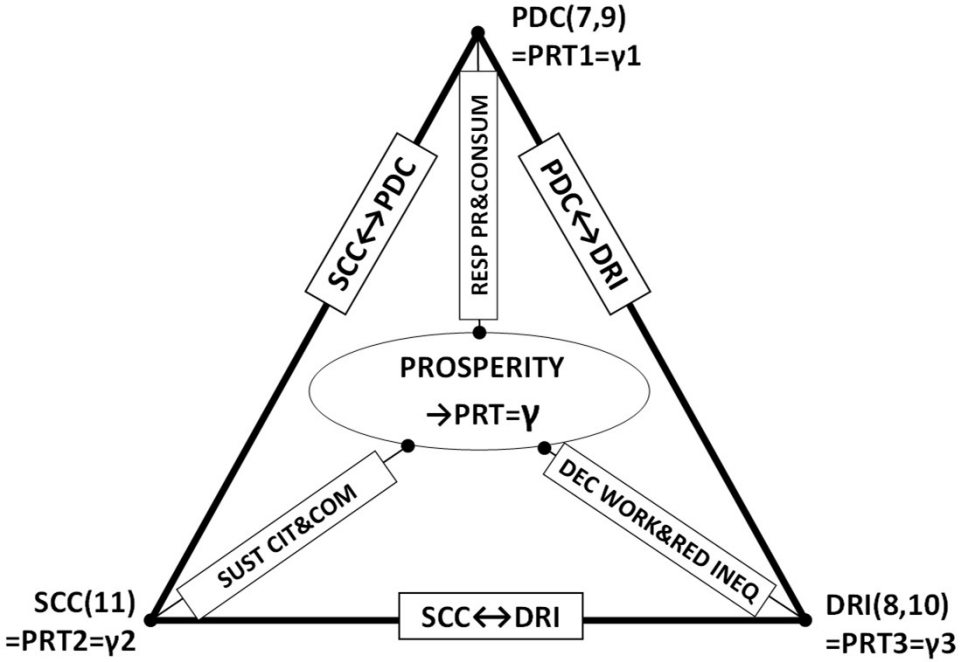
TRIÂNGULO SÍNTESE DE PESSOAS=PEOPLE (ODSs 1,2,3,4,5)

O TRIÂNGULO DA FIGURA MOSTRA A CONSOLIDAÇÃO DE P=PNT=PLANETA=PLANET=β (ODSs 6,12,13,14,15) EM TERMOS DE LIT(12,15)=PNT1=β1=LITHOSPHERE (ODSs 1,2); HYD(6,14)=PNT2=β2=HYDROSPHERE (ODSs 6,14); ATM(13)=PNT3=β3=ATMOSPHERE (ODS 13). ASSIM COMO NO TRIÂNGULO DA FIGURA "PEOPLE", A DIMENSÃO PLANET=PNT=β APARECE NA FACE DO TRIÂNGULO E AS ARESTAS DO TRIÂNGULO MOSTRAM OS RELACIONAMENTOS DIRECIONADOS ENTRE ESSES COMPONENTES.



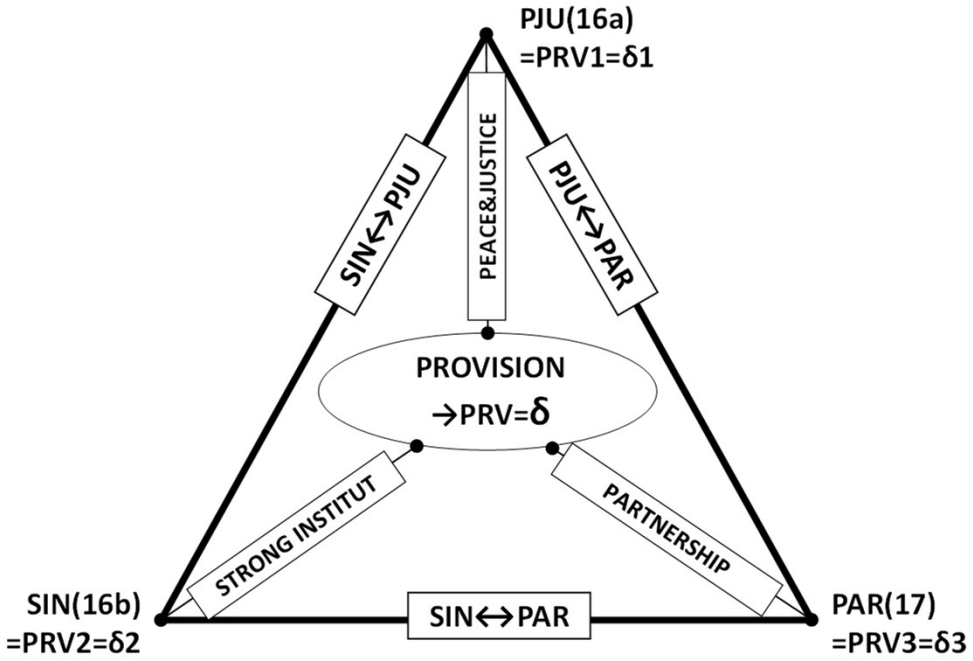
TRIÂNGULO SÍNTESE PLANETA=PLANET (ODSs 6,12,13,14,15)

O TRIÂNGULO DA FIGURA MOSTRA A CONSOLIDAÇÃO DE P=PRT=PROSPERIDADE=PROSPERITY=Y (ODSs 7,8,9,10,11) EM TERMOS DE PDC(7,9)=PRT1=Y1=PRODUCTION&CONSUMPTION (ODSs 7,9); SCC(11)=PRT2=Y2=SUSTAINABLE CITIES AND COMMUNITIES (ODS 11); DRI(8,10)=PRT3=Y3=DECENT WORK & REDUCE INEQUALITIES (ODSs 8,10). ASSIM COMO NOS TRIÂNGULOS DAS FIGURAS "PEOPLE" E "PLANET", A DIMENSÃO PROSPERITY=PRT=Y APARECE NA FACE DO TRIÂNGULO E AS ARESTAS DO TRIÂNGULO MOSTRAM OS RELACIONAMENTOS DIRECIONADOS ENTRE ESSES COMPONENTES.



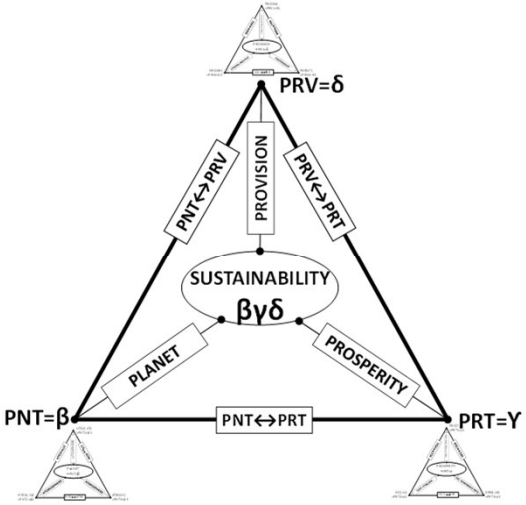
TRIÂNGULO SÍNTESE DE P=PROSPERIDADE=PROSPERITY (ODSs 7,8,9,10,11)

O TRIÂNGULO DA FIGURA MOSTRA A CONSOLIDAÇÃO DE P=PRV=PROVISÃO COM PARCERIA E PAZ=PROVISION WITH PARTNERSHIP AND PEACE=δ (ODSs 16,17) EM TERMOS DE PJU(16A)=PRV1=δ1=PEACE AND JUSTICE (ODS 16A); SIN(16B)=PRV2=δ2=STRONG INSTITUTIONS (ODS 16B); PAR(17)=PRV3=δ3=PARTNERSHIP (ODS 17). ASSIM COMO NOS TRIÂNGULOS DAS FIGURAS "PEOPLE", "PLANET" E "PROSPERITY", A DIMENSÃO PROVISION WITH PARTNERSHIP AND PEACE=PRV=δ APARECE NA FACE DO TRIÂNGULO E AS ARESTAS DO TRIÂNGULO MOSTRAM OS RELACIONAMENTOS DIRECIONADOS ENTRE ESSES COMPONENTES.

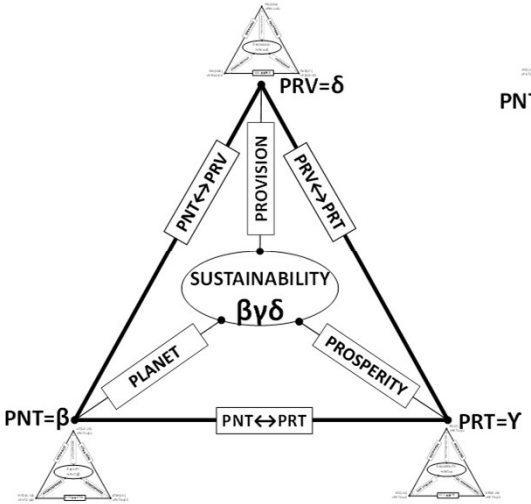


TRIÂNGULO SÍNTESE DE P=PROVISÃO COM PAZ E PARCERIA=PROVISION (ODSs 16,17)

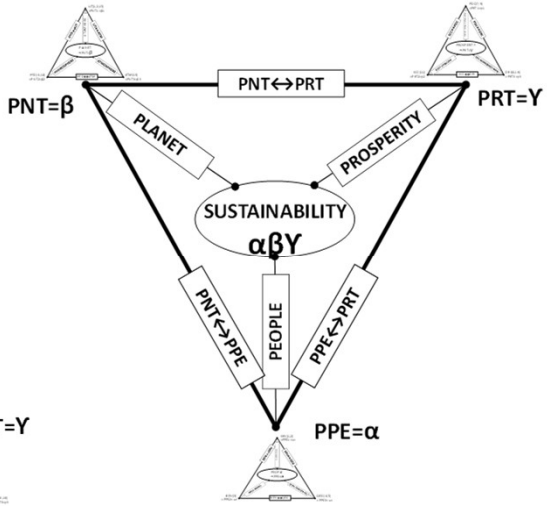
CADA UM DOS TRIÂNGULOS DAS FIGURAS EXPLICITAM, RESPECTIVAMENTE, A SÍNTESE DE 3 DAS 4 CONCENTRAÇÕES DE OBJETIVOS DE DESENVOLVIMENTO SUSTENTÁVEL=CONCENTRATIONS OF SUSTAINABLE DEVELOPMENT GOALS (P1=PESSOAS=PEOPLE=α; P2=PLANETA=PLANET=β; P3=PROSPERIDADE=PROSPERITY=Y; P4=PROVISÃO COM PARCERIA E PAZ=PROVISION WITH PARTNERSHIP AND PEACE=δ), EM QUE OS TRIÂNGULOS EM SEUS VÉRTICES REFEREM-SE AOS RESPECTIVOS COMPONENTES DAS FIGURAS "PEOPLE", "PLANET", "PR0SPERITY" E "PROVISION" ASSOCIADAS AO DESENVOLVIMENTO SUSTENTÁVEL.



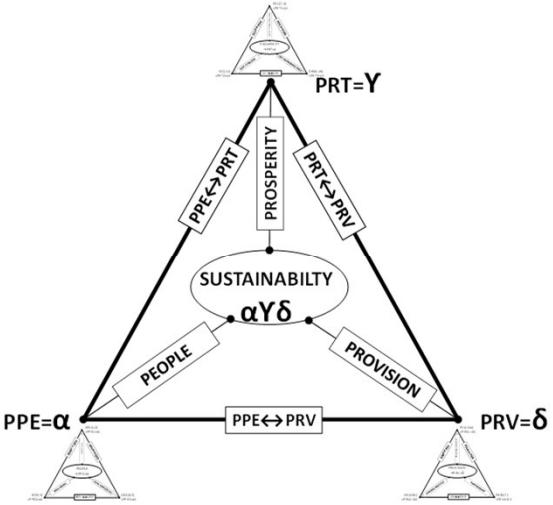
TRIÂNGULO SÍNTESE βγδ DESENVOLVIMENTO SUSTENTÁVEL



TRIÂNGULO SÍNTESE αβδ DESENVOLVIMENTO SUSTENTÁVEL



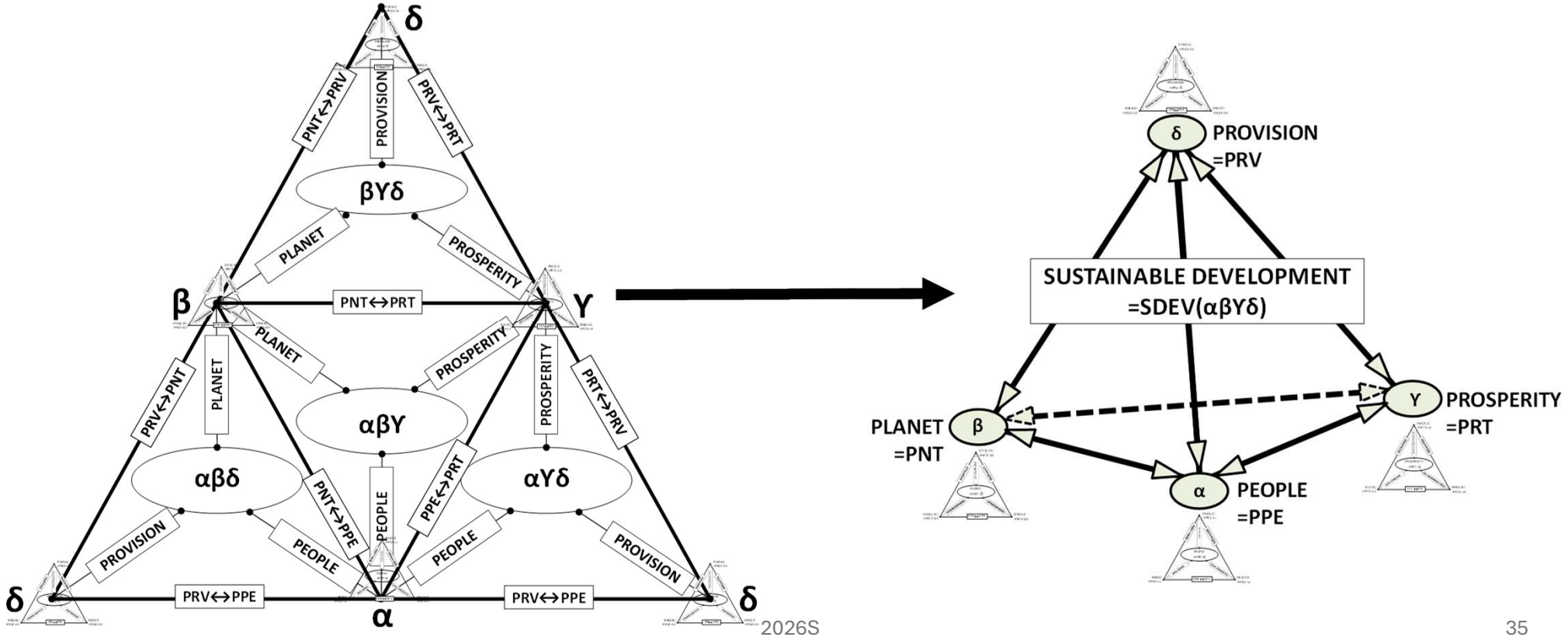
TRIÂNGULO SÍNTESE αβγ DESENVOLVIMENTO SUSTENTÁVEL



TRIÂNGULO SÍNTESE αγδ DESENVOLVIMENTO SUSTENTÁVEL

OBTÉM-SE O TRIÂNGULO DA FIGURA À ESQUERDA, QUE CONSTITUI UM TETRAEDRO PLANIFICADO, CONSTITUÍDO PELOS TRIÂNGULOS SÍNTESE (α,β,δ), (α,β,γ), (α,γ,δ) e (β,γ,δ).

O RESULTADO ESPERADO DO PROCESSO DE PREENCHIMENTO, VIA BIBLIOMETRIA, DO TETRAEDRO PLANIFICADO DA FIGURA À ESQUERDA É A SUA EFETIVA EVOLUÇÃO PARA UM TETRAEDRO ESPACIAL, COMO MOSTRA A FIGURA À DIREITA, QUE SINTETIZA AS BUSCAS BIBLIOMÉTRICAS ENVOLVENDO "PEOPLE", "PLANET", "PROSPERITY" E "PROVISION" EM DESENVOLVIMENTO SUSTENTÁVEL.



S1.4.1.1.3-BIBLIOGRAFIA ADICIONAL EM DESENVOLVIMENTO SUSTENTÁVEL

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MODELO BUSCAS BIBLIOMÉTRICAS: CONSOLIDA SUSTENTABILIDADE/INOVAÇÃO/PRIVACIDADE

COM BASE NOS TÓPICOS DESENVOLVIMENTO SUSTENTÁVEL (ARQUIVO 03.01IpgSipaSUSTENTABILIDADE...), PRONTIDÃO EM INOVAÇÃO (ARQUIVO 03.02IpgSipaINOVAÇÃO...) E PROTEÇÃO À PRIVACIDADE (ARQUIVO 03.03IpgSipaPRIVACIDADE...), PODE-SE CONSTRUIR O TRIÂNGULO DA FIGURA, QUE CONSOLIDA OS RELACIONAMENTOS SUSTENTABILIDADE, INOVAÇÃO E PRIVACIDADE EM UM TRIÂNGULO.

NOTAR AS CÓPIAS REDUZIDAS NOS VÉRTICES DO TRIÂNGULO DA FIGURA ASSOCIADAS À RESPECTIVA DIMENSÃO (SUSTENTABILIDADE, INOVAÇÃO E PRIVACIDADE).

